Artificial Intelligence Disclosure in the Annual Reports of Spanish IBEX-35 Companies (2018–2019)



Enrique Bonsón, Víctor Alejo, and Domenica Lavorato

Abstract This paper explores the information on Artificial Intelligence (AI) that Spanish IBEX 35 listed companies are including in their annual/sustainability reports. The study mainly focuses on the AI systems that companies indicate they are using or developing, the projects they are tackling and the extent to which they follow some principles or ethical guidelines when using AI-based technologies. The study analyses, both from a qualitative and quantitative perspective, the content of the reports of IBEX 35 companies for 2018 and 2019. The findings suggest that although AI reporting is growing because of the interest in these technologies, it is growing in a non-structured way, and that the adoption of ethical approaches to AI is at a very preliminary stage. The paper analyzes some evidence about a part of non-financial disclosure, AI disclosure, which has not been explored yet. It may serve as a starting point for researchers and companies interested in developing clear guidelines on what kind of information is relevant and mandatory for companies to report, what ethical principles or regulations AI applications must follow and how it has to be disclosed.

Keywords Artificial intelligence · Content analysis · Ethical guidelines · IBEX 35 companies · Non-financial reporting · Voluntary disclosure

1 Introduction

In April 2018, the European Commission declared its three main goals regarding artificial intelligence (AI): to increase the technological and industrial capacity of the European Union and the transfer of all those advances in AI to the economy; to prepare for the socio-economic changes brought about by AI; and, to ensure an appropriate ethical and legal framework [6]. This document led us to explore what

E. Bonsón (✉) · V. Alejo
University of Huelva, Calle Doctor Cantero Cuadrado, 6, Huelva, Spain
e-mail: bonson@uhu.es

D. Lavorato
Parthenope University of Naples, Via Amm. F. Acton, 38, 80133 Napoles, Italy

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2021

V. Kumar et al. (eds.), *Digital Transformation in Industry*, Lecture Notes in Information Systems and Organisation 44, https://doi.org/10.1007/978-3-030-73261-5_14

Spanish companies were doing in this respect or, more precisely, what they reported to do and what kind of risks and ethical approaches they considered [4].

In this study, which continues the one cited above, we analyze the information on artificial intelligence, machine learning, deep learning and big data, provided by Spanish IBEX-35 companies in their annual reports. We try to answer the following questions: Are companies reporting about artificial intelligence? In what kind of reports? Are they reporting only about general statements? What are the products or applications that these companies are developing and/or using? Have they launched AI projects and initiatives? Have they created or possess units or laboratories dedicated to their development? Do they analyze their potential risks? And finally, do they reflect on the ethical problems arising from their use? [4]. We also conduct a comparative analysis with the data from 2018 and 2019 to check how AI disclosure is evolving. This research constitutes a first approach to AI disclosure to gain insights to develop non-financial information guidelines related to artificial intelligence. These guidelines, should guarantee that the minimum requirements for trustworthy artificial intelligence, such as fairness, explainability, transparency and accountability, are met.

2 Literature Review

Although there is an extensive literature on sustainability reporting [1, 10, 11], the disclosure of information on artificial intelligence as an integral part of these reports has never been analyzed before and, therefore, there is no literature in that field which opens an interesting line of research.

Being a pioneer study in that line of research, we need to explore both current practices and targets to be achieved in terms of key indicators and reporting standards for AI. In that sense, some proposals have been launched on how to regulate artificial intelligence such as the European Commission's White Paper on Artificial Intelligence [7] or the Data Ethics Commission of the Federal Government of Germany [5], in which regulation is proposed to be based on the potential damage of an algorithmic system, addressing it from two fronts, the probability of damage occurring and its severity. On the other hand, we also consider some research that addresses the possibility of carrying out some standardization of all the proposals on ethical principles with "genuine mechanisms of accountability, external to companies and accessible to populations" [9] or some study on how start implementing ethics in AI and stop "AI ethics washing" [8].

3 Material and Methods

The methodology used for this study is that of content analysis. We downloaded all the annual/sustainability reports of these companies and looked into them for

mentions in artificial intelligence, machine learning, deep learning and big data. We got a total of 704 mentions that were then classified according to their content within the following categories: general statements, applications/uses, units/laboratories, risk and ethics.

As a "General statement", we consider all those mentions that do not address concrete AI topics, either talking about the future, either listing all these new technologies or talking about the changes that are taking place in business, among others. Here, for example, we expose what would be a typical mention of general matters: "The future of any company will rely on how they can lead in the adoption of Artificial Intelligence. At ArcelorMittal we are committed to being the leader in full enterprise digitalization" [2].

"Applications" include all those mentions explaining some specific developments or projects in which the company is involved and the products or applications they are using. Table 1 shows the main applications reported in 2018 annual reports, grouped into some of the most common functions.

As units/laboratories we imply all those mentions referring to a specific center, within the company, for the study and exclusive development of these new technologies.

In the risk category, we include all those companies that declare a possible negative impact that the development of artificial intelligence could cause. In our study, we found two companies that declare something about this, Bankinter [3], referring to the possible risk of theft of confidential big data, phishing or certain techniques that cybercriminals can develop to steal confidential data; and, Telefonica [13], referring to how they develop their products, applying a rigorous method both at the beginning and at the end of the design of their products and services to avoid possible adverse impacts on human rights.

Finally, our last category, closely related to the previous one, ethics, where we find nine of the 35 companies that make some kind of mention, highlighting the work of Telefónica [12], as one of the first companies to develop, in 2018, its own ethical principles, which include: fair AI, transparent and explainable AI, human-centric AI, privacy and security by design, and working with partners and third parties.

4 Results

The results of this study are presented in the following tables. Table 2 shows how many companies (%) report something related to each category. There is a considerable increase in all categories but the categories of applications and ethics stand out. This suggests that companies are increasingly using artificial intelligence and that they are slightly more concerned about the ethical aspects of the use of these powerful technologies. Appendix provides detailed company-by-company information.

On the other hand, Table 3 summarizes the level of AI disclosure in terms of the number of categories companies are reporting about.

Table 1 AI applications IBEX 35-2018

Category	Description	Company
Detection	Fraud	Amadeus
	Defects to enable predictive maintenance	ArcelorMIttal
	Money laundering and financing terrorism	B. Santander
	Cybersecurity threats	Bankia
	Money laundering and financing terrorism	BBVA
	Product safety issues and harmful substances	Inditex
	Threats to business operations	Telefónica
Diagnosis	Turbine health status	Acciona
	Diagnostic sensors in wind turbines	Siemens Gamesa
Optimization	Platform for massive data analysis	Bankia
	Cognitive platform to automate processes	Bankia
	Customer transaction inquiries	Bankia
	"Now", online banking app	CaixaBank
	Optimizing traffic congestion problems	Ferrovial
	Improving catalyst lifecycle reliability	Repsol
Prediction	Technical data management framework	Amadeus
	Detection of non-technical losses	Endesa
	Automatic report evaluation	Endesa
	Demand	Inditex
	Vehicle damage assessment	Mapfre
Recommendation system	Personalized offers for travelers	Amadeus
	Show offers from advertising partners	Amadeus
	Terminal recommendation	Telefónica
	Customer-based network services	Telefónica
Voice recognition	Chatbot	Amadeus
	Smart contact Center via voice identification	Bankinter
	Chatbot smart assistant	BBVA
	Chatbot	CaixaBank
	Call Center for cognitive assistance, home voice assist	CaixaBank
	Chatbot integrated into the social media platform	IAG
	Virtual Assistant Meliá	Meliá
	Aura digital assistant	Telfónica
Valuation	Personalized offers for travelers	Amadeus
	Show offers from advertising partners	Amadeus
	Terminal recommendation	Telefónica
	Customer-based network services	Telefónica

Table 2 Percentage of companies that report within each category

Category	IBEX 35-2018 (%)	IBEX 35-2019 (%)
General statement	66	77
Applications	63	77
Unit	9	14
Ethics	11	26
Risk	6	6

Table 3 AI disclosure (% of companies)

Type of disclosure	IBEX 35-2018 (%)	IBEX 35-2019 (%)
No disclosure	23	9
Just general statements	11	11
1–2 categories	49	49
3 categories	14	26
$\geq$ 4 categories	3	6

Table 4 AI reporting activity (% of companies)

	No activity (%)	Activity			Risk and ethic concerns (%)
		Low (%)	Medium (%)	High (%)	
IBEX 35 2018	34	49	14	3	17
IBEX 35 2019	20	49	26	6	32

It shows clear evidence that companies stop having so many worthless mentions, and begin to declare more about specific issues and give more information about the uses they give to these new technologies.

Finally, Table 4 shows the level of reporting activity and its progress, it is worth noting that companies that either did not have mentions or only disclosed general matters have dropped by 14%, which means an increase in that amount in AI Reporting Activity. Highlighting the concern of companies for ethical and risk aspects. That is consistent with the results shown in the tables above.

5 Discussion

This study is a first approach to what companies report in their annual and sustainability reports on artificial intelligence, machine learning, deep learning and big data. It is a first analysis where we check how current disclosure is and we begin to think on how it should be, what issues companies should report about to clarify which products/services are using these technologies, in what aspects they are influencing

the company and what possible uses are being given. Some standards on what and how companies have to report AI related matters in their non-financial information sections of their annual reports should be developed to guarantee both transparency and accountability for the stakeholders.

6 Conclusions, Limitations and Future Research

According to the results of this preliminary research we conclude the following: (1) AI reporting activity is growing as AI is becoming more used in companies, (2) It is growing in a non-structured way, (3) It will become more important to have a specific sub-section on artificial intelligence in the non-financial information section of the annual reports. (4) There is a need for clear guidelines on what information is relevant and mandatory for companies to report and what ethical principles or regulations these AI applications must comply with so that they can be carried out without negative implications for society.

A series of limitations should be recognized in this research, the most important of which would be that we focus only on the IBEX 35 companies, when we should cover a wide range of companies to obtain a more global sample. In addition, our methodology needs to be automated in order to carry out a more complete and efficient study. Even so, it is a good starting point for researchers and companies interested in digging deeper.

Appendix. AI Disclosure by Company

	General statement		Application		Unit/Lab		Risk/Impact		Ethic	
Company	2018	2019	2018	2019	2018	2019	2018	2019	2018	2019
ACCIONA	1	1	1	1	0	0	0	0	0	0
ACERINOX	1	1	1	1	0	0	0	0	0	0
ACS	1	1	0	1	0	0	0	0	0	0
AENA	–	–	–	–	–	–	–	–	–	–
AMADEUS	1	1	1	0	1	0	0	0	0	0
ARCELORMITTAL	1	1	1	1	0	0	0	0	0	0
B. SABADELL	–	1	–	0	–	0	–	0	–	1
B. SANTANDER	1	1	1	1	0	0	0	0	1	1
BANKIA	1	1	1	1	0	0	0	0	0	1
BANKINTER	1	1	1	1	0	0	1	1	0	0
BBVA	1	1	1	1	0	0	0	0	1	1
CAIXABANK	1	1	1	1	0	0	0	0	0	1
CELLNEX	1	1	0	0	0	0	0	0	0	0
CIE AUTOMOTIVE	1	1	0	0	0	0	0	0	0	0
ENAGAS	0	0	1	1	0	0	0	0	0	0
ENCE	–	0	–	1	–	0	–	0	–	0
ENDESA	1	1	1	1	0	0	0	0	0	1
FERROVIAL	1	1	1	1	0	0	0	0	0	0
GRIFOLS	–	–	–	–	–	–	–	–	–	–
IAG	0	1	1	1	0	1	0	0	0	0
IBERDROLA	1	1	1	1	0	0	0	0	0	0
INDITEX	1	1	1	1	0	0	0	0	0	0

(continued)

(continued)

	General statement		Application		Unit/Lab		Risk/Impact		Ethic	
Company	2018	2019	2018	2019	2018	2019	2018	2019	2018	2019
INDRA	1	1	0	1	0	0	0	0	1	1
INM COLONIAL	–	1	–	1	–	0	–	0	–	0
MAPFRE	0	0	1	1	0	0	0	0	0	0
MASMOVIL	0	1	1	0	0	0	0	0	0	0
MEDIASET	1	1	1	1	0	0	0	0	0	0
MELIA	1	1	1	1	0	0	0	0	0	0
MERLIN	–	0	–	1	–	0	–	0	–	0
NATURGY	1	1	1	1	0	0	0	0	0	0
RED ELECTRICA	1	1	0	1	0	1	0	0	0	1
REPSOL	1	1	1	1	1	1	0	0	0	0
SIEMENS GAMESA	–	0	–	1	–	1	–	0	–	0
TELEFONICA	1	1	1	1	1	1	1	1	1	1
VISCOFAN	–	–	–	–	–	–	–	–	–	–
	23	27	22	27	3	5	2	2	4	9

References

1. Al-Shaer, H. (2020). Sustainability reporting quality and post-audit financial reporting quality: Empirical evidence from the UK. *Business Strategy and the Environment, 29*(6), 2355–2373. https://doi.org/10.1002/bse.2507
2. ArcelorMittal. (2019). Integrated report. Retrieved from https://corporate.arcelormittal.com/sustainability.
3. Bankinter. 2019. Annual report. Retrieved from https://webcorporativa.bankinter.com/www2/corporativa/en/inf_financiera_cnmv/informacion_financiera/memoria/2019.
4. Bonsón, E., & Lavorato, D. (2019). La inteligencia artificial en las empresas del IBEX 35. *Revista AECA, 128*, 127–129
5. Data Ethics Commission of the Federal Government (2019). *Opinion of the data ethics commission.* Retrieved from www.datenethikkommission.de.
6. European Commission (2018). *Artificial intelligence. A European perspective.* Retrieved from https://ec.europa.eu/jrc/en/publication/artificial-intelligence-european-perspective.
7. European Commission (2020). White Paper—On Artificial Intelligence—A European approach to excellence and trust. Brussels.
8. Hao, K. (2019). *In 2020, let's stop ai ethics-washing and actually do something.* Retrieved from https://www.technologyreview.com/2019/12/27/57/ai-ethics-washing-time-to-act/.
9. Hickok, M. (2020). Lessons learned from AI ethics principles for future actions. *AI and Ethics, 1*(1), 41. https://doi.org/10.1007/s43681-020-00008-1.
10. Richarson, A., & Kachler, M. (2017). University sustainability reporting: a review of the literature and development of a model. In J. Arevalo, S. Mitchell (Eds.), *Handbook of sustainability in management education* (pp. 385–405). https://doi.org/10.4337/9781785361241.00027.
11. Romero, S., Ruiz, S., & Fernandez-Feijoo, B. (2018). Sustainability reporting and stakeholder engagement in Spain: Different instruments, different quality. *Business Strategy and the Environment, 28*(1), 221–232. https://doi.org/10.1002/bse.2251
12. Telefónica (2018). Nuestros principios de Inteligencia Artificial. Retrieved from https://www.telefonica.com/es/web/negocio-responsable/nuestros-compromisos/principios-ia.
13. Telefónica (2019). Integrated report. Retrieved from https://www.telefonica.com/en/web/about_telefonica/publications/annual-reports.